LECTURE 1.3 · SHIP AND OCEAN SYSTEMS DESIGN

The Design Process

Papanikolaou (2026) §1.3.1–1.3.4 and §1.3.6.1; Tables 1.5–1.6 for Exercise 1.3 · Session W1.2, second half · Lecture 20 min + Exercise 1.3 20 min + Activity 1.3 10 min

Luís Batista

Instituto Superior Técnico / University of Lisbon

luis@veranavis.com

calendly.com/vnsd-luisbatista/meetme

The Life of a Ship: from Concept to Recycling

1

Pre-Contractual Work

Long negotiation and study before signing.

Most ships are built to the owner's needs.

Series from standard designs are growing.

2

Design and Engineering

Basic design approved by class and flag.

Detail and production drawings follow.

Equipment is ordered meanwhile.

3

Production

Steel is cut and formed.

Blocks are built, outfitted and painted.

Blocks are joined in the dock.

4

Launching

The hull goes into the water.

A milestone for payment and schedule.

Outfitting continues afloat.

5

Commissioning and Delivery

Systems are tested at the quay.

Sea trials prove speed and manoeuvring.

The owner accepts the ship.

6

Operation and Maintenance

Twenty to thirty years in service.

Class surveys and dry-dockings.

Repairs, upgrades and conversions.

7

Ship Recycling

Sold at the end of its life.

Dismantled at a recycling yard.

Steel and equipment are recovered.

The Actors

Owner side

Owner	Orders and owns
Manager	Operates, maintains
Charterer	Hires the ship
Cargo owner	Provides the cargo
Site team	Supervises the build

Finance and legal

Banks	Finance, guarantees
Export credit	State-backed loans
Insurers	Cover the risks
Broker	Order and sale deals
Lawyers	Draft the contract

Crew

Master and crew	Sail and maintain
Crewing agency	Recruits, trains crew

Design

Consultant	Concept, advice
Design office	Basic design
Detail design	Production drawings
Model basin	Tank tests

Build

Shipyard	Builds and delivers
Sub-contractors	Blocks, systems
Repair yard	Docking, repairs

Suppliers and services

OEMs	Engines, equipment
OEM service	Spares, after-sales
Steel suppliers	Plates and profiles
Bunker suppliers	Supply the fuel
Salvors	Emergency response

Rules and control

IMO	Sets conventions
Regional bodies	EU, coastal rules
Flag state / RO	Certificates
Port state	Inspects in port
Class approval	Approves drawings
Class survey	Surveys the build
Class in service	Periodical surveys

Ports

Port authority	Runs the port
Pilots and tugs	Bring the ship in
Terminals	Load and discharge
Ship agent	Handles port calls

End of life

Cash buyer	Buys for recycling
Recycling yard	Dismantles the ship

From the Design Basis to a First Design Loop

1

The process

Four phases, the design spiral and why design iterates

2

The first loop

Read the brief critically, gather data on similar ships, list the rules

E

Exercise 1.3

Which design phase made each document of the reference ship? Then Activity 1.3: its real documents

Read: Papanikolaou (2026) §1.3.1–1.3.4 and §1.3.6.1; Tables 1.5–1.6

Bring: 0.6 The Reference Ship, slides 2 and 3

Sequential Engineering: the Design Spiral

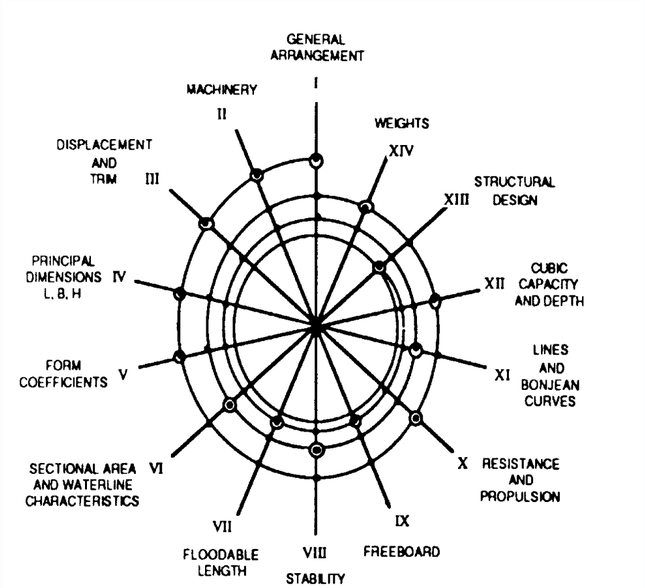


Fig. 3 Evan's general design diagram [40]

Evans (1959): the design spiral, a sequential and iterative process.

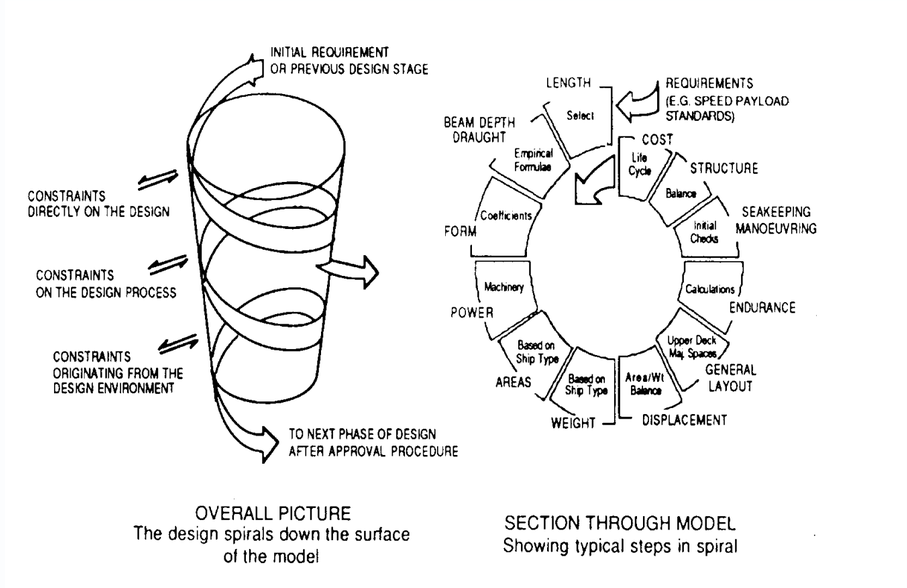


Fig. 4 Andrew's overall model of the ship design process [42]

Buxton (1972) added the economic aspects; Andrews (1981) added time as a third dimension.

Main Stages of Ship Design

I. Concept, Basic and Detail	**Concept**	**Basic**	**Detail**		
II. Classical four stages	**Concept**	**Preliminary**	**Contract**	**Detail**	
III. Five stages	**Concept**	**Preliminary**	**Contract**	**Detail**	**Production**

Common to all definitions

Mission Requirements	**Basic Design Plans & Calculations for Classification Society**	**Statutory Design Plans & Calculations for Flag**	**Detail Design for Production**

The level of design depends on

Type of ship commercial or military, for example	**Client** owner, shipyard or another designer	**Shipyard** technological processes and documentation requirements	**Classification society** class notation
Manpower expertise skills and experience of the Production team	**Complexity** size, systems and novelty of the ship	**Integrated Logistic Support** spares, manuals and training	**Design workflow process** tools, data exchange and approvals

Different Perspectives on Design

1. By Scope / Level of Detail	Concept Design	Basic Design	Detail Design	As-Built / Construction Design		
2. By Design Phase / Project Stage	Feasibility Study	Preliminary Design	Functional Design	Production design	Delivery & Commissioning	
3. By Process Flow	Front-End Design (FED)	Engineering Design	Production Design	Integration & Verification	Support & Lifecycle	
4. By Discipline / Specialisation	Naval Architecture	Marine Engineering	Structural Engineering	Electrical & Control Systems	HVAC / Utilities	Outfitting
5. By Type of Vessel	Commercial	Defence	Specialised	High-Speed / Advanced	Green / Alternative	
6. By Client / Stakeholder	Shipowner / Operator	Shipyard	Classification Society	Flag State / Regulatory Bodies	Design Office / Engineering Firm	
7. By Tools / Technology	Traditional CAD	3D Modelling	CAE / FEA	CFD	PLM	Digital Twin
8. By Organisational Approach	In-house / Full Design Office	Outsourced / Partnered	Consortium / Joint Venture	Alliance / New Company		

Example of application

1 · Concept

Ship type · Deadweight · Propulsion system · Service speed

2 · Preliminary

General arrangement sketch · Preliminary lines plan · Propulsive power · Lightship weight · Initial cost

3 · Contract

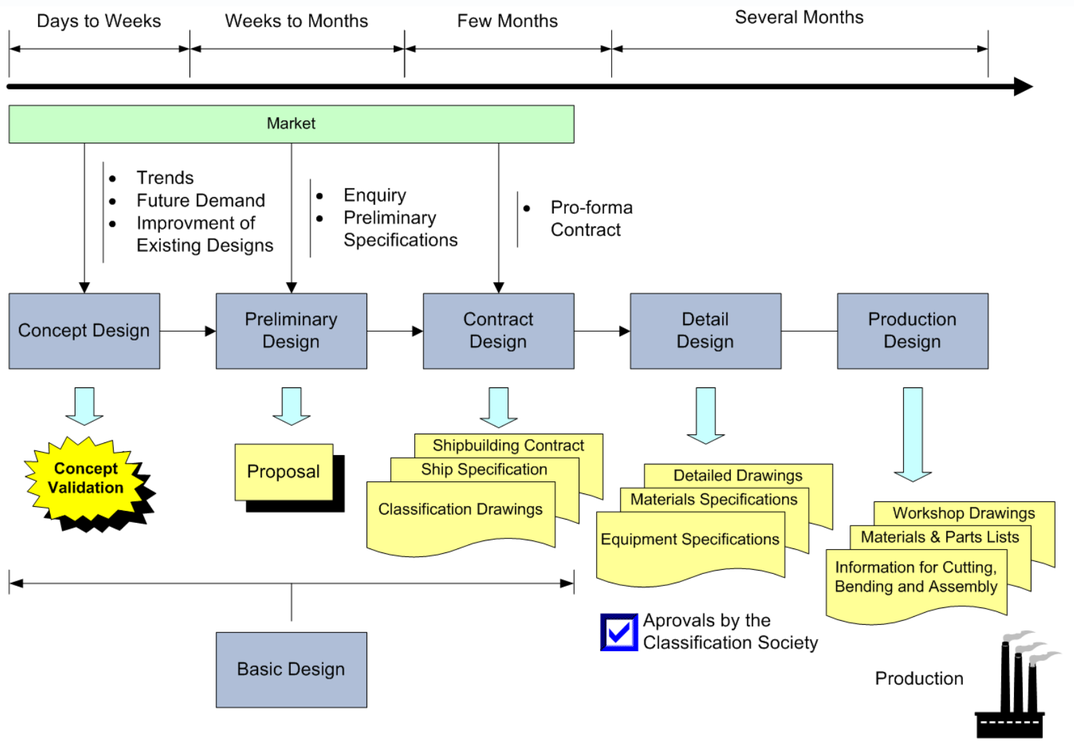
Specification and general arrangement · Lines plan for tank tests · Stability and structural computations · Some classification drawings · Single-line piping diagrams

4 · Detail

Faired lines or 3D hull model · Detailed general arrangement · All classification drawings · Building blocks · Construction planning · Piping and duct arrangements

5 · Production

Production drawings, to block level · Material and parts lists · Cutting, bending and assembly data · Piping isometrics



The stages along the time scale

Four Phases, Each a Turn of the Spiral

BASIC DESIGN · THIS COURSE · D1–D10

A · Concept

1st loop · feasibility

Starts from the owner’s brief. Ship type, deadweight, propulsion, speed; a first L, B, D, T, CB and PB; alternatives explored.

Today: about a day

B · Preliminary

2nd to 4th loop · optimum

Main dimensions, hull form, machinery, arrangement, structure, stability, freeboard, tonnage. GA sketch, power, lightship, cost.

Effort: about 15 × A

C · Contract

5th loop · specification

Faired lines and tank tests, structural and stability computations. The specification (machinery, class) and general arrangement signed with the contract.

Effort: about 17 × B

D · Detailed

Production

Detail and production (slide 8, stages 4 and 5): 3D model, class drawings, every detail, parts lists and workshop data.

Effort: the largest by far

After A + B the yard can price the ship and tender. These early decisions fix most of the cost of the ship’s **whole life**, from building to recycling.

Today: the Whole Loop in One Model

Mission

Route · Capacity · Speed · Restrictions

Function

Payload: cargo spaces, handling, treatment. Ship: machinery, tanks, crew, structure

Form

Main dimensions · Hull lines · Space and weight balance · Stability

Performance

Resistance · Propulsion · Structure · Safety · Emissions

Economics

Building cost · Operating cost · Required freight rate

A **synthesis model** computes all five views of a design variant at once, so thousands of variants can be compared (HOLISHIP, Harries and Abt 2019). The steps run in parallel, but the loop remains: **every change is recalculated** until the design closes.

> Owner, yard, class, ports and cargo owners pull in different directions, so even the first loop is an optimization with conflicting goals. Your D3 Ship Model is this synthesis model.

The First Loop Starts with the Brief

1

Evaluate

Which requirements fix the main dimensions? Which are vague, missing or in conflict?

2

Gather data

Similar ships: type, size, DWT or TEU, speed, power. D2; next session and App. A.

3

List the rules

SOLAS, ICLL, MARPOL, class rules for this type and size; ECA and port rules on the route.

Questions the 0.2 Ship Design Project briefs leave open

Is the TEU capacity nominal or at a given weight per TEU? · At what draught and engine rating is the speed? · What are the draught and length limits at each port? · How many days of reserve beyond one round voyage or loop?

EXERCISE 1.3 · INDIVIDUAL · 20 MIN · PRACTICE FOR D1

Which Design Phase Made This Document?

Eight documents of the real ship behind **the reference ship**. Phases of slide 9: **A Concept · B Preliminary · C Contract · D Detailed**.

	Document
a	Owner’s brief: deadweight, speed, route (0.6, slide 2)
b	First estimate of displacement and main dimensions
c	Principal particulars: LBP, beam, depth, design and scantling draughts
d	Main engine: make, type and rating
e	Class notation: +KRS1 Oil/Chemical Tanker … Product/II (IBC)
f	Capacity plan: 14 cargo tanks, volumes and centres of gravity
g	Lines plan for the model tests
h	Cutting and assembly data for the steel blocks

1 Match · 8 min

For each document, the phase in which it is first produced (or “before A”).

2 What came first · 8 min

For d, e and f: one decision that had to be taken before that document could be written.

Check: the worked answer is shown at the end (Exercise Answers – Week 1). Not graded.

ACTIVITY 1.3 · WHOLE CLASS · 10 MIN

The Reference Ship’s Documents

Six kinds of document describe a ship. The real ship behind **the reference ship** has all of them; the next slide shows two.

1 Specifications

What the ship must be and have

Owner’s brief (0.6, slide 2) · building specification · engine maker’s specification

2 Drawings

Where everything is

General arrangement · capacity plan · lines plan · midship section

3 Calculations

Proof that it works

Trim and stability booklet · longitudinal strength · power prediction and model tests · EEDI file

4 Data sheets and lists

The ship in numbers

Principal particulars · tank table · lightweight list · makers list

5 Certificates

Approved by class and flag

Class certificate with the KR notation (0.6, slide 3) · IOPP · Certificate of Fitness (IBC) · IEE

6 Production data

How to build it

Block division · cutting and nesting data · assembly sequence

For each document shown: 1 Which kind is it? · 2 Who writes it, and who approves it (slide 3)? · 3 When is it used again in the ship’s life (slide 2)?

Same Ship, Two Documents

4 Data sheet · Principal particulars

Item	Yard sheet
LOA · LBP	hidden
Breadth · depth, moulded	hidden
Design draught, moulded	9.80 m
Scantling draught	hidden
Block coefficient	hidden
Displacement · lightship	hidden
Deadweight, summer	37,596 t
Main engine	MAN B&W ME-B type; model and ratings hidden
Service speed	about 14.5 kn at the design draught, NCR, 15 % sea margin
Class	KR, notation as 0.6, slide 3
Flag · freeboard	Marshall Islands · type A
Complement	25 + 6 P

2 Drawing · Capacity plan, cargo tanks

Tanks	Frames	100 %, m³	98 %, m³
14 cargo tanks, No. 1–7 P/S	46–89	39,794.9	38,999.0
2 slop tanks P/S	43–46	2,674.4	2,620.9
Total		**42,469.3**	**41,619.8**

The plan also gives each tank’s centre of gravity, for the stability booklet and the loading instrument.

> The data sheet sums the ship up for owners, brokers and charterers; the capacity plan is used by the crew on every voyage.

Extracts retyped from the yard’s documents. **Hidden** values are answers to Exercises 1.4 and 1.5 and to week 2.

The Story in Two Lines

1 · Process. Basic design is the first four turns of the spiral; a synthesis model now runs each turn, but the loop remains.

2 · First loop. Read the brief critically, find similar ships, list the rules, then size the ship.

Next, Fri 6 Nov (W1.3): design methods, first displacement and main dimensions. Read §1.3.6–1.3.7 and §2.1–2.2.